CRE-seq: Explicit encoding of transcription initiation grammar improves regulatory sequence generation

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Abstract

Massively parallel reporter assays (MPRAs) and sequence-to-activity models have enabled de novo design of cis-regulatory elements (CREs) by optimizing predicted activity directly from DNA sequence. However, most current design pipelines treat transcription initiation grammar—experimentally characterized constraints on motif position, orientation, and combinatorial organization—as an implicit property of a surrogate model, limiting mechanistic control and interpretability. Here we present CRE-seq, a syntax-aware CRE generator that explicitly encodes promoter initiation rules derived from high-resolution nascent transcription studies, and we use CRE-seq to test a focused hypothesis: under an otherwise identical optimization and evaluation pipeline, do explicit grammar constraints improve sequence generation beyond unconstrained surrogate-guided search?

CRE-seq performs genetic-algorithm (GA) optimization of fixed-length promoter-like sequences using an MPRA-trained activity predictor, and optionally applies binary penalties for violations of canonical initiation grammar (e.g., TATA/Inr/downstream element windows and spacing constraints). To isolate the effect of grammar, baseline and grammar-aware configurations share the same search procedure, scoring model, and evaluation metric. Using PARM-K562 as a unified evaluator, we find that both CRE-seq configurations achieve predicted activities in the extreme right tail of an external reference distribution of previously published machine-designed sequences (CODA), providing an interpretable activity-scale context. Critically, adding explicit grammar penalties yields a systematic right-shift in predicted activity relative to baseline CRE-seq (one-sided Wilcoxon rank-sum test, $n = 50$ per group, $p = 1.1 \times 10^{-13}$), with 92% of baseline outcomes falling below the grammar-aware median. Together, these results demonstrate that explicit transcription initiation grammar provides actionable design

constraints that can complement strong sequence-to-activity surrogates, improving regulatory sequence generation while preserving mechanistically interpretable promoter architecture.

Introduction

Precise control of gene expression depends on cis-regulatory elements (CREs), including promoters and enhancers, which determine where and how transcription is initiated. Early sequence models emphasized the presence or absence of individual transcription factor (TF) motifs, yet regulatory activity also depends on higher-order organization: motif spacing, orientation, and combinatorial interactions can strongly modulate transcriptional output, motivating the concept of regulatory "grammar" beyond single-motif effects (White et al., 2013; Shlyueva et al., 2014).

Nascent transcription assays have refined our understanding of initiation architecture at nucleotide resolution. Measurements of newly synthesized RNA indicate that initiation occurs across constrained regions rather than at a single fixed nucleotide, and promoters and enhancers share a common initiation organization (Core et al., 2014; Danko et al., 2015; Andersson et al., 2017; Henriques et al., 2018). Within this framework, canonical promoter elements exhibit position-dependent contributions and context sensitivity, suggesting that interpretable constraints on motif placement and spacing can be derived from experimental data.

In parallel, MPRA datasets have enabled sequence-to-activity predictors that accurately map DNA sequence to regulatory output (Kwasnieski et al., 2014; Bogard et al., 2023). Recent work has coupled such predictors to optimization or generative procedures to design new regulatory sequences, including cell-type-targeting CREs (Gomes et al., 2024). These approaches are powerful but are typically implemented as black-box search over sequence space, with motif grammar treated as an implicit property learned by the model.

This raises a practical and testable question for CRE engineering: when a strong surrogate predictor already captures much of the sequence–activity relationship, does explicitly enforcing experimentally inferred initiation grammar still provide additional value for sequence generation?

Here, we introduce CRE-seq, a syntax-aware CRE generator that explicitly encodes transcription initiation rules as design constraints during sequence optimization. Rather than attempting to learn grammar from scratch, CRE-seq operationalizes promoter syntax reported in high-resolution initiation studies as concrete penalties applied during search. Importantly, we use CRE-seq to perform a controlled comparison: baseline and grammar-aware configurations share identical optimization settings and the same activity model, differing only by the presence of grammar constraints. This design isolates the contribution of explicit grammar to the optimization process and provides a mechanistic, reproducible strategy for improving CRE design beyond purely black-box approaches.

Methods

Overview of the CRE-seq design framework

CRE-seq is a genetic algorithm (GA)–based framework for generating fixed-length promoter-like sequences with high predicted activity while optionally enforcing explicit transcription initiation grammar constraints. Candidate sequences are iteratively optimized through selection, mutation, and crossover, guided by a fitness function that combines a surrogate-predicted activity score with additive penalties for grammar violations (grammar-aware configuration).

Optimization surrogate and unified evaluation model

CRE-seq uses an MPRA-trained sequence-to-activity model as the optimization surrogate. For reporting results and enabling consistent comparisons across sequence sets, all sequences generated in this study, as well as external reference sequences, are evaluated using a unified evaluation model, PARM-K562.

Because the same model family (PARM) is used during optimization and evaluation, we restrict scientific inference to comparisons that hold the surrogate model fixed. The primary comparison is therefore baseline versus grammar-aware CRE-seq, which share identical optimization procedures and evaluation settings and differ only in the inclusion of grammar penalties.

Genetic algorithm optimization

Unless otherwise stated, each GA run uses a population size of 64 sequences and is optimized for 80 generations. Initial populations are generated by sampling each nucleotide position uniformly from {A, C, G, T}.

At each generation, candidate sequences are scored by the optimization surrogate and selected using tournament selection (tournament size 3). Elitism retains the single highest-scoring sequence from the current generation. New candidate sequences are generated via per-base mutation and single-point crossover. Mutation is applied independently at each nucleotide position with probability 0.01. Crossover events occur with probability 0.5, with crossover points sampled uniformly along the sequence length.

All stochastic components use fixed random seeds (default seed = 1) to ensure reproducibility across runs.

CRE-seq configurations

We report results for three sequence sets. CODA sequences are a previously published set of machine-designed regulatory sequences (Gomes et al., 2024), included here solely as an external reference for predicted activity scale. CODA sequences were optimized using a different surrogate model and objectives; accordingly, we do not treat CODA as a direct benchmark for method superiority. For scale contextualization, CODA sequences are rescored using the PARM-K562 evaluation model.

Baseline CRE-seq sequences are generated using the GA optimization described above without grammar penalties. Grammar-aware CRE-seq sequences are generated using the same optimization procedure, genetic operators, and scoring model, but with additional grammar-based penalties in the fitness function. This design ensures that observed differences between baseline and grammar-aware CRE-seq arise solely from explicit grammar constraints.

Motif-based grammar constraints

Motifs are represented using IUPAC-degenerate consensus patterns and are detected via exact matching (not PWM scoring). Motifs are scanned within predefined windows relative to a fixed transcription start site (TSS) coordinate system, where the TSS is defined as the center of the sequence.

TATA-like motifs (TATAWAWR) are scanned in an upstream window spanning positions −33 to −26 relative to the TSS. Initiator-like motifs (YYANWYY) are scanned in a narrow window centered on the TSS. Downstream promoter motifs (RGWYV) are scanned in a downstream window spanning positions +28 to +33 relative to the TSS. Binary penalties are applied when expected motifs are absent from their corresponding windows.

Syntax constraints and spacing rules

In addition to motif presence, CRE-seq encodes spacing relationships between promoter elements following core promoter syntax characterized at single-nucleotide resolution (Danko et al., 2024). When both initiator and downstream motifs are present, their relative spacing is required to fall within a predefined range; violations incur a binary penalty.

Fitness function

For grammar-aware CRE-seq, fitness is computed as the surrogate-predicted activity score minus additive penalties for grammar violations. Motif and syntax penalties are binary indicators and are weighted equally. Baseline CRE-seq omits penalty terms and uses only the surrogate-predicted activity score.

Activity-scale contextualization using CODA

To contextualize the absolute scale of predicted activities achieved by CRE-seq, we summarize the distribution of CODA scores under PARM-K562 using empirical cumulative distribution functions (ECDFs). CRE-seq median scores are overlaid as reference thresholds to indicate where CRE-seq outcomes lie relative to the external reference distribution.

Results

CRE-seq achieves high predicted activity relative to an external reference scale

To contextualize predicted activity magnitudes under a unified evaluator, we rescored a published set of machine-designed sequences (CODA) with PARM-K562 and treated the resulting distribution as an external reference scale (Fig. 1). Under PARM-K562, CODA scores concentrate in a modest range and approach saturation at relatively low values, providing a practical reference distribution for previously published machine-guided designs under the same metric.

Both baseline and grammar-aware CRE-seq runs achieve median predicted activities that lie in the extreme right tail of the CODA distribution (Fig. 1). In our experiments, essentially none of the rescored CODA sequences reach or exceed the CRE-seq median, indicating that CRE-seq operates in a high-activity regime that is well separated from the bulk of the external reference distribution under the same evaluation model.

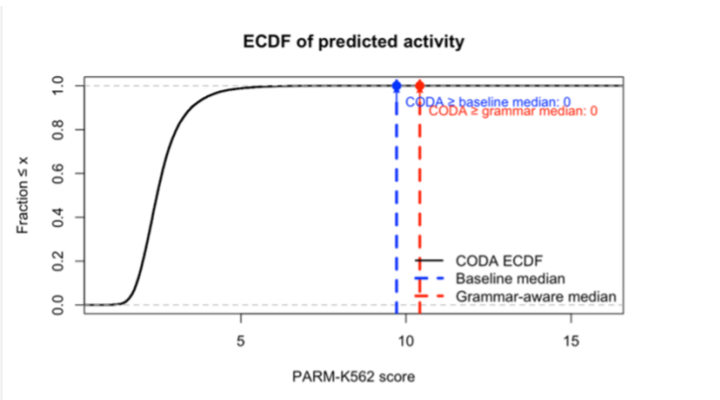

Figure 1

Explicit grammar constraints improve generation under controlled optimization settings

To isolate the effect of explicit grammar, we compared baseline and grammar-aware CRE-seq under identical GA settings, identical surrogate scoring, and identical evaluation (PARM-K562). Adding grammar penalties yields a systematic right-shift in predicted activity relative to baseline CRE-seq while enforcing promoter-like initiation syntax by construction (Fig. 2).

Quantitatively, 92% of baseline outcomes fall below the grammar-aware median, consistent with a broad improvement rather than a small number of outliers. The best-scoring sequences achieved PARM-K562 scores of 11.72 (baseline) and 14.01 (grammar-aware). Grammar-aware CRE-seq achieves significantly higher predicted activity than baseline CRE-seq (one-sided Wilcoxon rank-sum test, $n = 50$ per group, $p = 1.1 \times 10^{-13}$).

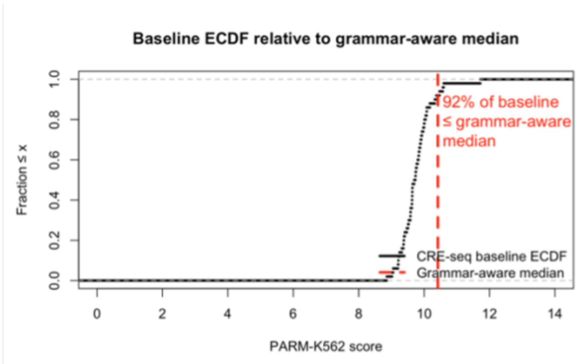

Figure 2

Discussion

In this study, we used a controlled sequence generation framework to test whether explicitly encoding experimentally inferred transcription initiation grammar provides additional value beyond a strong MPRA-trained surrogate model. By holding the optimization procedure, scoring model, and evaluation metric fixed, we isolated the effect of grammar constraints on sequence generation. Under this controlled setup, grammar-aware CRE-seq consistently achieved higher predicted activity than an unconstrained baseline, indicating that explicit promoter syntax can contribute actionable information during optimization even when using a powerful sequence-to-activity predictor.

Rescored CODA sequences served as an external reference distribution for interpreting predicted activity magnitudes under a unified evaluator. Because CODA sequences were generated using different surrogates and objectives, we do not interpret differences between CODA and CRE-seq as method superiority. Instead, the CODA ECDF provides a practical activity-scale context, demonstrating that CRE-seq operates in a high-activity regime relative to previously published machine-designed sequences when evaluated under the same model.

A common intuition is that accurate sequence-to-activity models implicitly capture regulatory grammar, rendering explicit constraints unnecessary. Our results suggest that this intuition is incomplete in the context of generative optimization. In CRE-seq, grammar penalties restrict exploration toward sequences consistent with experimentally supported initiation architecture, guiding the search away from degenerate solutions that achieve high predicted scores while violating canonical promoter organization. Explicit grammar thus acts as a complementary inductive bias, improving optimization outcomes while preserving mechanistically interpretable sequence structure.

This study is computational and relies on a single MPRA-trained surrogate model and a restricted promoter grammar implemented in one cellular context (K562). Accordingly, our conclusions are limited to the value of explicit grammar constraints under a fixed predictor rather than claims about biological activity in vivo. Extending this framework to additional cell types, alternative surrogates, and a broader set of experimentally supported grammar rules, as well as experimental validation of generated sequences, represents an important direction for future work.

Data and code availability

All code required to reproduce the analyses and generate sequences is available in the CRE-seq GitHub repository (see https://github.com/kjkjdhy/CRE-seq). No new experimental datasets were generated for this study.

Competing interests

The author declares no competing interests.

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Author contributions

H.D. conceived the study, implemented the CRE-seq framework, performed analyses, and wrote the manuscript.

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